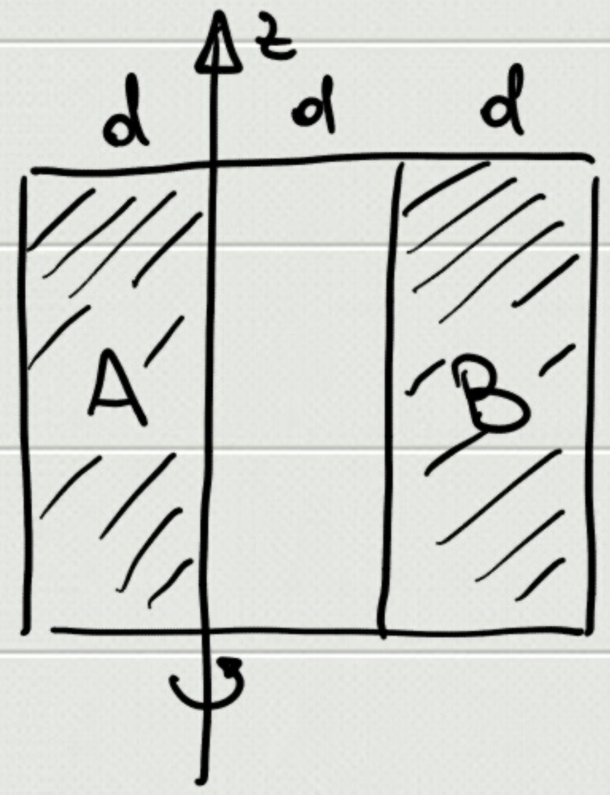


$d = 0.24\ m$

$m_A = m = 11\ kg$

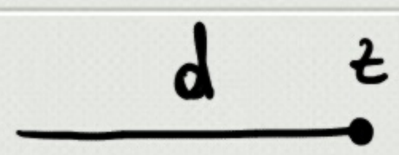
$m_B = 2m_A$

$\omega_0 = 0 \qquad \omega_1 = 8\ rad/s$

$W_{0\to1} = ?$

$W = \int M\,d\theta \qquad W_{TOT} = \Delta E_k = E_{k,fin} - \emptyset$

$W_{0\to1} = \frac{1}{2} I_z \omega_1^2 \qquad I_z = I_A + I_B$



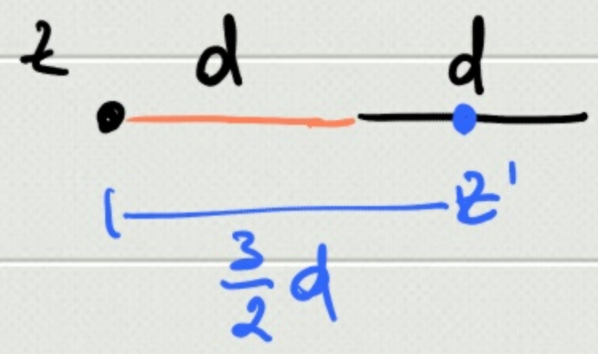
$I_A = \frac{1}{3} m_A d^2$

$I_B = \cancel{\frac{1}{3} m_B d^2} + m_B d^2$

$I_B = \cancel{\frac{1}{12} m_B d^2} + m_B d^2$

$I_B = \frac{1}{12} m_B d^2 + m_B \left(\frac{3}{2} d\right)^2 \quad *$

$I_B = \cancel{m_B \left(\frac{3}{2} d\right)^2}$



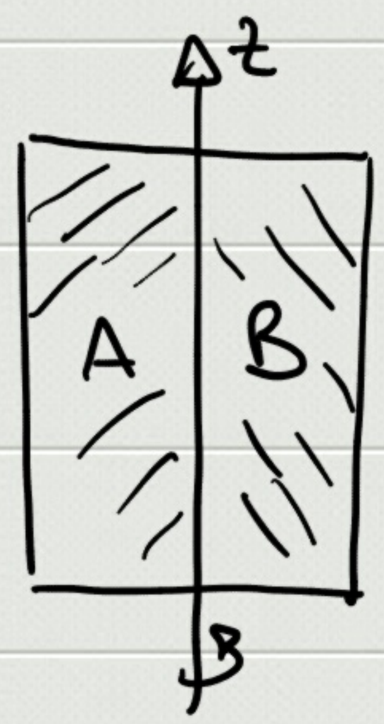
$I_z = I_{z'} + m a^2$

$$I_A = \frac{1}{3} m d^2$$

$$I_B = \frac{1}{\cancel{12}_{6}} \cancel{2} m d^2 + \cancel{2} m \frac{9}{\cancel{4}_{2}} d^2$$

$$I_z = I_A + I_B = m d^2 \left(\frac{1}{3} + \frac{1}{6} + \frac{9}{2} \right) = 5 m d^2$$

$$W_{0 \to 1} = \frac{1}{2} I_z \omega_1^2 = \frac{1}{2} 5 m d^2 \omega_1^2 = 101.4 \ J$$

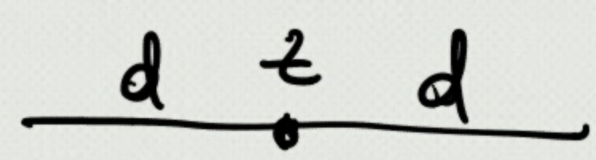


$$W_{1 \to 2} = ?$$

$$W_{1 \to 2} = \Delta E_k = \frac{1}{2} I_z' \omega_2^2 - \frac{1}{2} I_z \omega_1^2$$

$$I_z' = \frac{1}{12} \cancel{(m_A + m_B)} d^2$$

$$I_z' = \frac{1}{12} \cancel{(m_A + m_B)} (2d)^2$$



$$I_z' = \frac{1}{3} m_A d^2 + \frac{1}{3} m_B d^2 \quad * \quad = m d^2$$

$$I_z' = \left[\frac{1}{12} m_A d^2 + m_A \left(\frac{d}{2} \right)^2 \right] + \left[\frac{1}{12} m_B d^2 + m_B \left(\frac{d}{2} \right)^2 \right]$$

*

$$\bar{M}_0^{\varepsilon} = 0 \Rightarrow \boxed{\bar{L} = \text{cost}}$$

$$\Rightarrow \bar{L}_1 = \bar{L}_2 \Rightarrow \bar{I}_z \omega_1 = \bar{I}_z' \omega_2$$

$$\Rightarrow \omega_2 = \frac{I_z}{I_z'} \omega_1 = \frac{5md^2}{md^2} \omega_1 = 5\omega_1$$

$$\Rightarrow W_{1\to 2} = \frac{1}{2} I_z' \omega_2^2 - \frac{1}{2} I_z \omega_1^2 = 10 md^2 \omega_1^2 =$$

$$= 405.5 \text{ J}$$